

Wavefunction-based methods

Post-Hartree-Fock

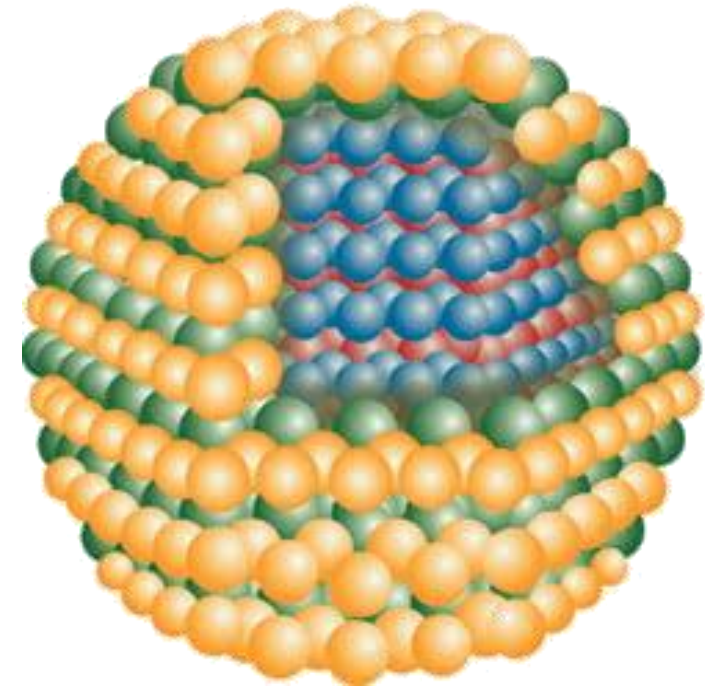
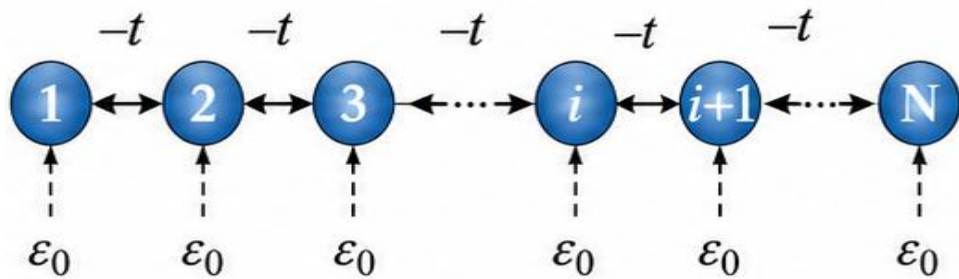
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Electronic Structure

From model systems to real structures and materials



The many-body problem

Schrödinger equation for N particles:

$$\hat{H}\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = E\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$$

Time-dependent version:

$$\hat{H}\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t) = i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t)$$

Hamiltonian:

$$\hat{H} = \sum_{i=1}^N \left[-\frac{\hbar^2}{2m_i} \nabla_i^2 + V_{\text{ext}}(\mathbf{r}_i) \right] + \sum_{i<j}^N V_{\text{int}}(\mathbf{r}_i, \mathbf{r}_j)$$

N	$\dim(\mathcal{H}_N)$
1	2
2	4
3	8
10	1024
20	1,048,576
30	1,073,741,824

Hartree-Fock in a nutshell

Electronic Hamiltonian

$$\hat{H} = \sum_{i=1}^N \hat{h}(i) + \sum_{i<j} \frac{1}{r_{ij}}, \quad \hat{h}(i) = -\frac{1}{2} \nabla_i^2 - \sum_A \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|}$$

Wavefunction:

$$\Psi_{\text{HF}}(1, \dots, N) = \frac{1}{\sqrt{N!}} \det[\phi_i(x_j)]$$

Task:

$$E_{\text{HF}} = \min_{\{\phi_i\}} \langle \Psi_{\text{HF}} | \hat{H} | \Psi_{\text{HF}} \rangle$$

Equation to solve:

$$\hat{F} \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}), \quad \langle \phi_i | \phi_j \rangle = \delta_{ij}$$

Fock operator:

$$F_{pq} = h_{pq} + \sum_{r,s} P_{rs} [(pq|rs) - \frac{1}{2}(pr|qs)]$$

$$J_{pq} = \sum_{r,s} P_{rs} (pq|rs) \quad \text{Coulomb interactions}$$

$$K_{pq} = \sum_{r,s} P_{rs} (pr|qs) \quad \text{Exchange interactions}$$

Total Energy

$$E_{\text{HF}} = \sum_{p,q} P_{pq} h_{pq} + \frac{1}{2} \sum_{p,q} P_{pq} [J_{pq} - K_{pq}]$$

E.g., H vs. He

```
srun -p labs --reservation=abinitio ./script.py
```

```
from pyscf import gto, scf

mol = gto.M(
    atom = 'H 0 0 0',
    basis = 'cc-PVDZ',
    charge = 0,
    spin = 1
)

mf = scf.UHF(mol)
energia_hf = mf.kernel()
```

HF Energy = ?

Exact Energy = ?

```
from pyscf import gto, scf

mol = gto.M(
    atom = 'He 0 0 0',
    basis = 'cc-PVDZ',
    charge = 0,
    spin = 0
)

mf = scf.RHF(mol)
energia_hf = mf.kernel()
```

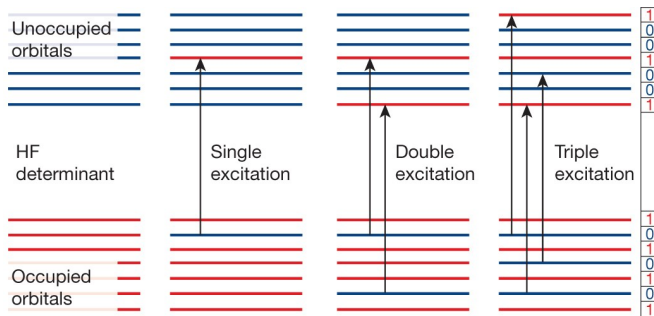
HF Energy = ?

Exact Energy = ?

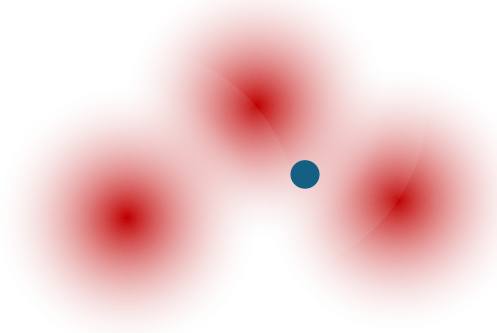
What is wrong with **Hartree-Fock**?

$$\text{e-e Many-Body Interactions} = \underbrace{\text{Hartree-Fock}}_{\text{Hartree-Fock}} + \text{Higher-order diagrams} + \dots$$

Single-Determinant Ansatz



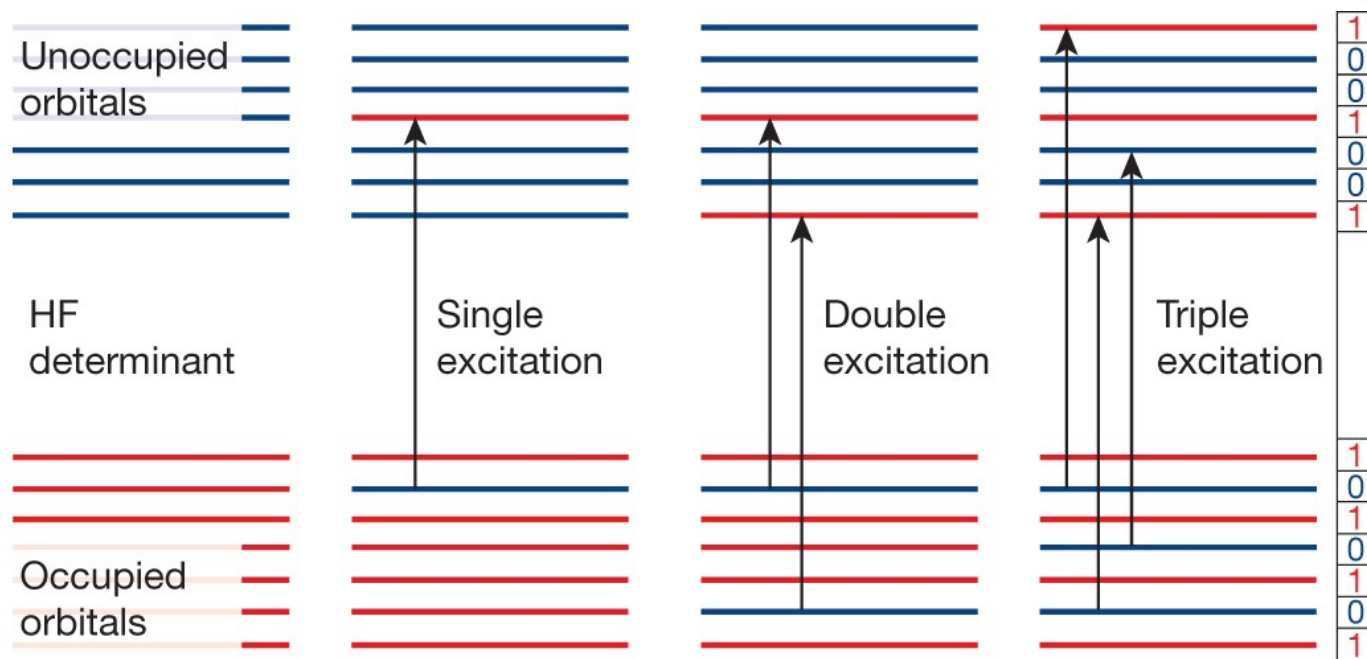
Mean-field approximation



Others

- Born-Oppenheimer approximation
- Finite basis set
- Spin restriction/contamination
- Neglect relativistic and QED effects

Improving Hartree-Fock: Configuration Interaction



$|\Phi_0\rangle$: HF reference determinant

$|\Phi_i^a\rangle = a_a^\dagger a_i |\Phi_0\rangle$: Single excitations

$|\Phi_{ij}^{ab}\rangle$: Double excitations

Task:

Determine the CI coefficients by solving a new secular equation

Still variational!

$$|\Psi_{\text{CI}}\rangle = c_0 |\Phi_0\rangle + \sum_{i,a} c_i^a |\Phi_i^a\rangle + \sum_{i<j, a<b} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle + \dots$$

But...

	$ \Phi_0\rangle$	$ \Phi_i^a\rangle$	$ \Phi_{ij}^{ab}\rangle$	$ \Phi_{ijk}^{abc}\rangle$
$ \Phi_0\rangle$				
$ \Phi_i^a\rangle$		Dense		
$ \Phi_{ij}^{ab}\rangle$			Sparse	
$ \Phi_{ijk}^{abc}\rangle$				

CISD

Very Sparse

Level	Scaling
CIS	$O(N^4)$
CISD	$O(N^6)$
CISDT	$O(N^8)$
FCI	$O(e^N)$

CISD (or any other truncation) has problems...

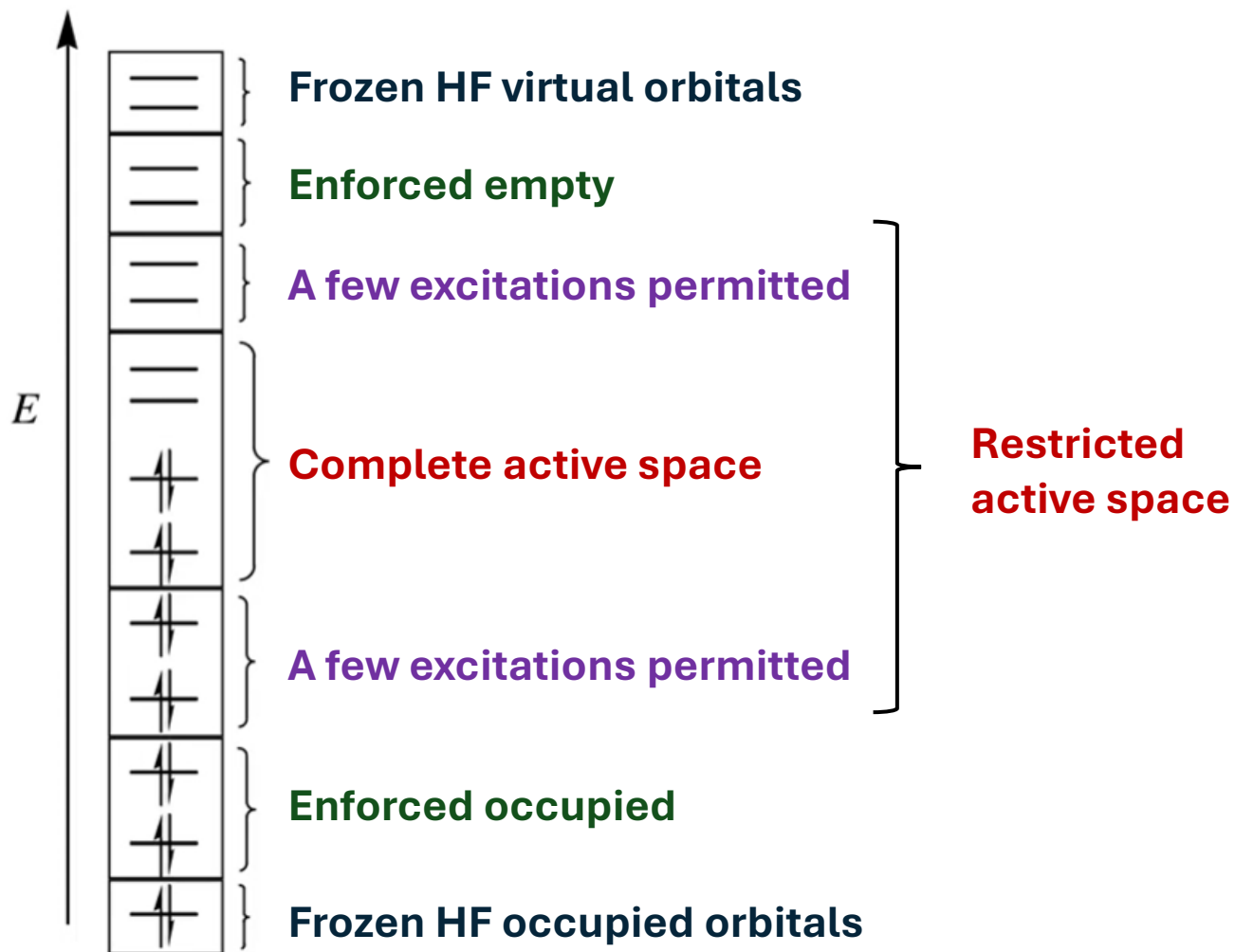


$$\begin{aligned}\Psi_{\text{CI}} &= \mathcal{A}\Psi_{\text{CI(a)}}\Psi_{\text{CI(b)}} \\ &= \mathcal{A}(a_0|\sigma^2\rangle_a + a_2|\sigma^{*2}\rangle_a)(a_0|\sigma^2\rangle_b + a_2|\sigma^{*2}\rangle_b) \\ &= \mathcal{A}\left(a_0^2|\sigma^2\rangle_a|\sigma^2\rangle_b + a_0a_2|\sigma^{*2}\rangle_a|\sigma^2\rangle_b + a_0a_2|\sigma^2\rangle_a|\sigma^{*2}\rangle_b + \cancel{a_2^2|\sigma^{*2}\rangle_a|\sigma^{*2}\rangle_b}\right)\end{aligned}$$

Violation of size-consistency

More alternatives:

Multiconfiguration self-consistent field (MCSCF)



When not doing FCI:

- ☐ Reoptimize (relax) the coefficients of the most important orbitals
- ☐ More excitations into lower-energy orbitals

E.g.

CASSCF

Complete active space SCF

RASSCF

Restricted active space SCF

Coupled-Cluster to the rescue

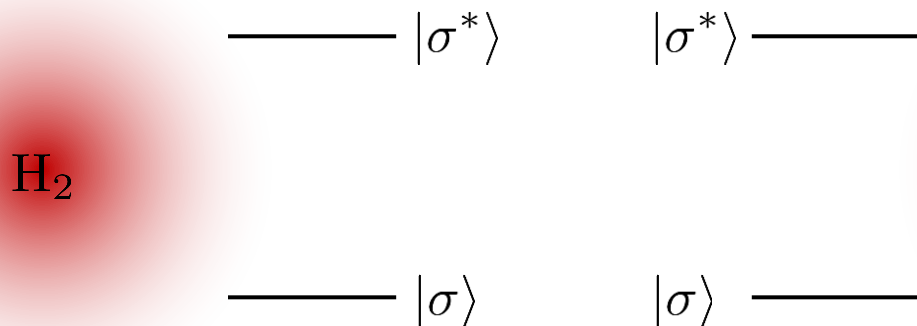
$$|\Psi_{\text{CC}}\rangle = e^{\hat{T}} |\Phi_0\rangle, \quad \hat{T} = \hat{T}_1 + \hat{T}_2 + \hat{T}_3 + \dots$$

$$\hat{T}_1 = \sum_{i,a} t_i^a a_a^\dagger a_i, \quad \hat{T}_2 = \frac{1}{4} \sum_{ij,ab} t_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i, \dots$$

```
from pyscf import cc

ccsd_calc = cc.CCSD(mf)
E_corr_ccsd, t1, t2 = ccsd_calc.kernel()
print("Energía CCSD:", E_corr_ccsd + energia_hf)
print("Correlación CCSD:", E_corr_ccsd)
```

→ Try a CCSD calculation



Does it conserve size-consistency?

Scaling of Coupled-Cluster

Method	Ansatz	Amplitudes included	Scaling	Property
CCD	$\hat{T} \approx \hat{T}_2$	Doubles only (no T_1)	$O(N^6)$	size-extensive
CCSD	$\hat{T} \approx T_1 + T_2$	Singles + Doubles	$O(N^6)$	size-extensive
CCSD(T)	CCSD + perturbative (T)	CCSD amplitudes + T_3 estimate	$O(N^7)$	"gold standard"
CCSDT	$T_1 + T_2 + T_3$	up to triples	$O(N^8)$	size-extensive

Comparison between CC and CI

Property	CISD	CCSD	CCSD(T)
Size-Consistency	X	✓	✓
Size-Extensivity	X	✓	✓
Recovery of Triples	X	~ via T_1T_2	✓ perturbatively
Variational	✓ (upper bound)	X	X
Accuracy/Cost	$O(N^6)$	$O(N^6)$	$O(N^7)$

A different approach: **Perturbation theory**

Møller–Plesset (MP) perturbation theory

$$\hat{H} = \hat{H}^{(0)} + \lambda \hat{V}, \quad \lambda \in [0, 1]$$

$$\hat{H}^{(0)} = \sum_{i=1}^N \hat{F}(i) \quad \hat{V} = \hat{H} - \hat{H}^{(0)}$$

Series expansion:

$$E = E^{(0)} + \lambda E^{(1)} + \lambda^2 E^{(2)} + \lambda^3 E^{(3)} + \dots$$

$$|\Psi\rangle = |\Psi^{(0)}\rangle + \lambda |\Psi^{(1)}\rangle + \lambda^2 |\Psi^{(2)}\rangle + \dots$$

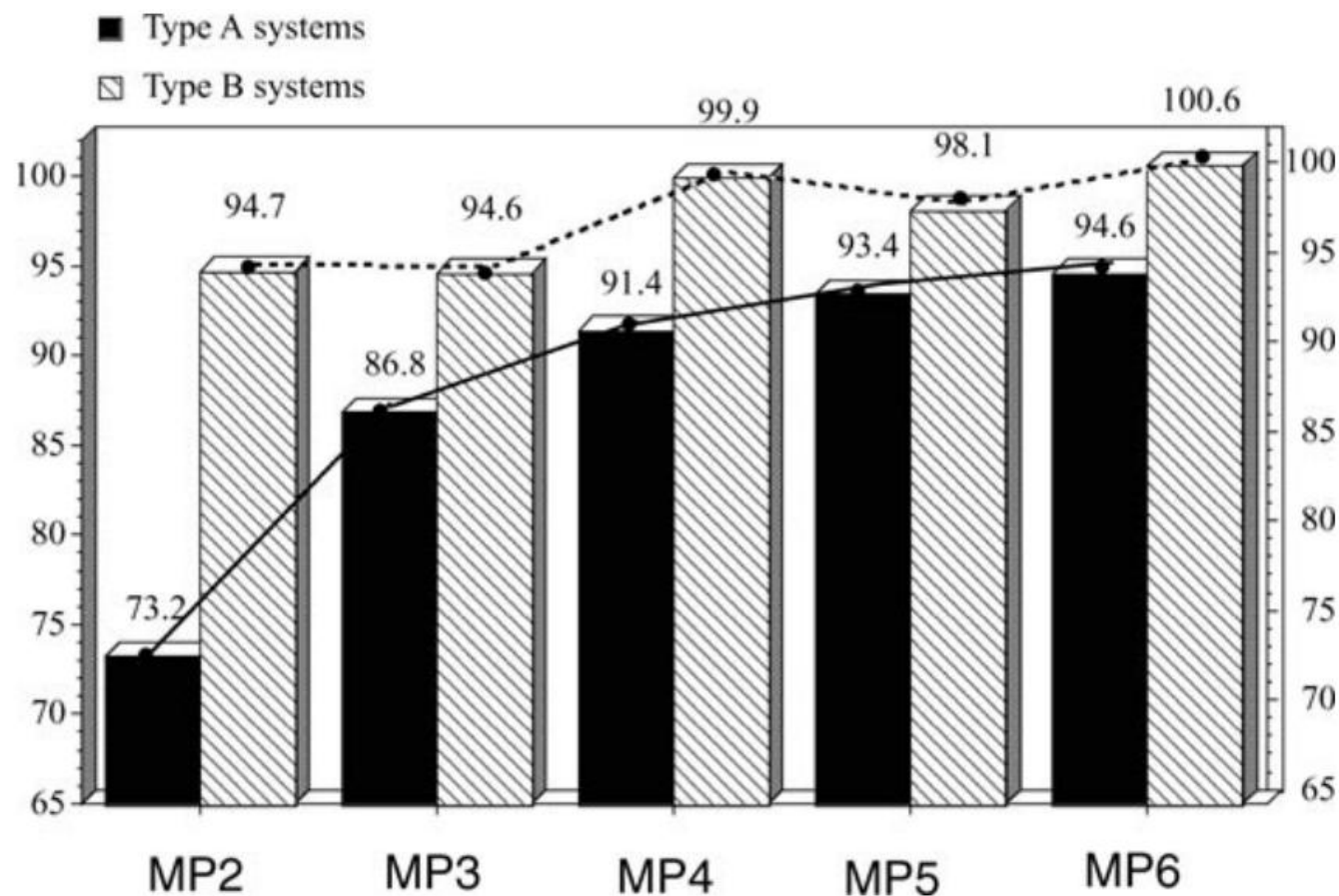
$$E^{(0)} = \langle \Phi_0 | \hat{H}^{(0)} | \Phi_0 \rangle = \sum_i^{\text{occ}} \varepsilon_i$$

$$\begin{aligned} E^{(1)} &= \langle \Phi_0 | \hat{V} | \Phi_0 \rangle = \langle \Phi_0 | \hat{H} - \sum_i \hat{F}(i) | \Phi_0 \rangle \\ &= E_{\text{HF}} - \sum_i \varepsilon_i \end{aligned}$$

$$E^{(2)} = \sum_{I \neq 0} \frac{|\langle \Phi_0 | \hat{V} | \Phi_I \rangle|^2}{E^{(0)} - E_I^{(0)}} = \frac{1}{4} \sum_{ijab} \frac{\langle ij || ab \rangle^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}$$

$$E^{(3)} = \sum_{I \neq 0} \sum_{J \neq 0} \frac{\langle \Phi_0 | \hat{V} | \Phi_I \rangle \langle \Phi_I | \hat{V} | \Phi_J \rangle \langle \Phi_J | \hat{V} | \Phi_0 \rangle}{(E^{(0)} - E_I^{(0)})(E^{(0)} - E_J^{(0)})}$$

MPn Performance



Type A: Uniformly distributed electronic density

Type B: Presence of highly electronegative atoms

Average MPn correlation energies given in percent of the FCI correlation energy for 14 class A and 19 class B examples.

J Phys Chem 1996, 100:6173–6188

Build your own MP2

```
from pyscf import ao2mo

# Obtener integrales ERI en base MO, notación químico (pq|rs)
nmo = mf.mo_coeff.shape[1]
eri_1d = ao2mo.kernel(mol, mf.mo_coeff)
eri_mo = ao2mo.restore(1, eri_1d, nmo) # tensor 4D (p,q,r,s)

eps = mf.mo_energy
nocc = mol.nelectron // 2

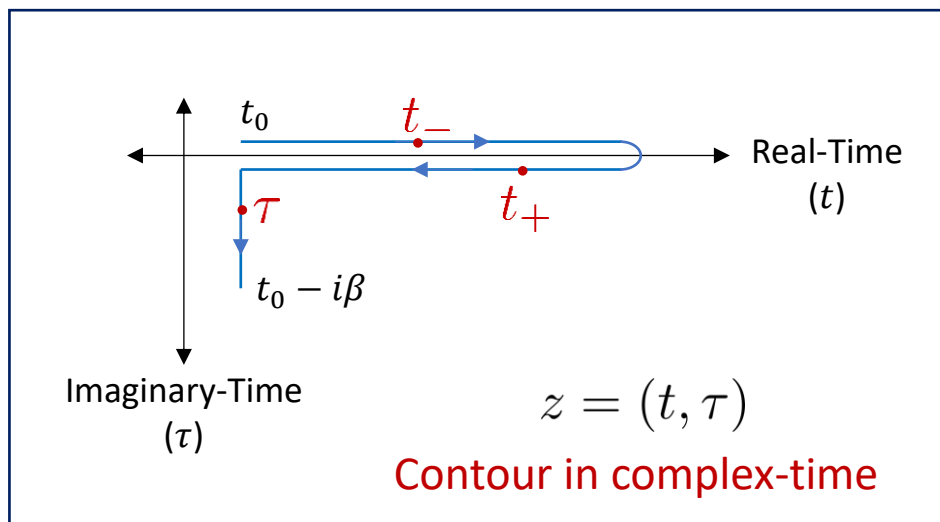
MP2_corr_E = ? #Completar script
```

Simplified expression for restricted MP2

$$E_{\text{corr}}^{\text{MP2}} = \sum_{iajb} \frac{(ia|jb) [2(ia|jb) - (ib|ja)]}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}$$

- $(ij|kl)$ Electron repulsion integrals
- $\{ij\}$ Occupied orbitals
- $\{ab\}$ Virtual orbitals

Green's functions: A completely different perspective



One-particle Green's function

$$G_{ij}(z, z') = -\frac{i}{\hbar} \left\langle \mathcal{T}_c \left\{ \hat{a}_i(z) \hat{a}_j^\dagger(z') \right\} \right\rangle \longrightarrow \text{Ensemble average}$$

Contour (time)
ordering operator

Creation and
annihilation operators

When projected to the real or imaginary time axes:

Matsubara GF

$$G^M(\tau, \tau') \equiv G(t_0 - i\tau, t_0 - i\tau')$$

Greater GF

$$G^>(t, t') \equiv G(t_+, t'_-)$$

Left GF

$$G^\lceil(\tau, t) \equiv G(t_0 - i\tau, t_\pm)$$

Retarded GF

Lesser GF

$$G^<(t, t') \equiv G(t_-, t'_+)$$

Advanced GF

Right GF

$$G^\rceil(\tau, t) \equiv G(t_\pm, t_0 - i\tau)$$

Physical information contained in the **Green's function**

Time-ordered single-particle Green's function in position representation:

$$i\hbar G(rt, r't') = \langle \Psi_H | \mathcal{T}[\hat{\Psi}_H(rt) \hat{\Psi}_H^\dagger(r't')] | \Psi_H \rangle \quad \left\{ \begin{array}{ll} \langle \Psi_H | \hat{\Psi}_H(rt) \hat{\Psi}_H^\dagger(r't') | \Psi_H \rangle & t' < t \\ -\langle \Psi_H | \hat{\Psi}_H^\dagger(r't') \hat{\Psi}_H(rt) | \Psi_H \rangle & t < t' \end{array} \right.$$

Many-body state in
Heisenberg's representation

Creation field operator

For $t > t'$

$$i\hbar G(rt, r't') = \langle \Psi_S(t) | \hat{\Psi}_S(r) e^{-i\hat{H}(t-t')/\hbar} \hat{\Psi}_S^\dagger(r') | \Psi_S(t') \rangle$$

Using the Schrödinger representations

Heisenberg (H) and Schrödinger (S) representations

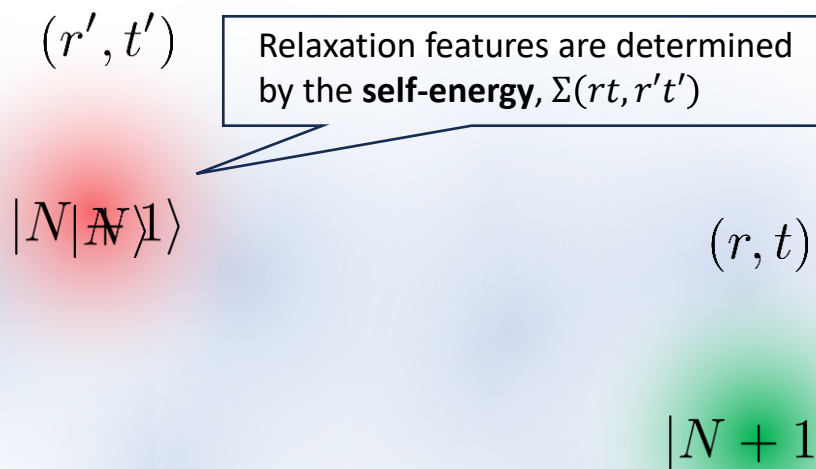
$$|\Psi_H\rangle = e^{i\hat{H}t/\hbar} |\Psi_S(t)\rangle$$

$$\hat{O}_H(rt) = e^{i\hat{H}t/\hbar} \hat{O}_S(r) e^{-i\hat{H}/\hbar}$$

Physical information contained in **GF**

The “electron channel”

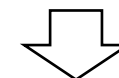
$$i\hbar G(rt, r't') = \theta(t - t') \langle \Psi_S^0(t) | \hat{\Psi}_S(r) e^{-i\hat{H}(t-t')/\hbar} \hat{\Psi}_S^\dagger(r') | \Psi_S^0(t') \rangle$$



Adds an electron at r' to the N -electron ground state



Propagates the $(N+1)$ -state from time t' to t



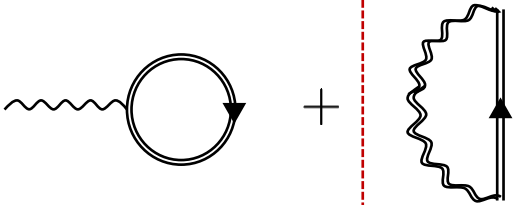
Projects the propagated state into a state with an added electron at r, t

The electron channel of the GF can be interpreted as the propagator of the added electron

Common approximations for the **self-energy**

GW Closure: First-order approximation in the screened Coulomb interactions

$$\Sigma^{\text{GW}} = \text{Hartree-Fock} + \text{GW diagram}$$

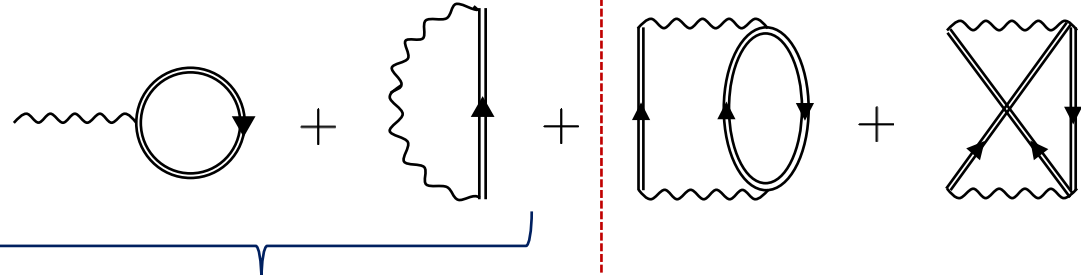
$$\Sigma_{ij}(t, t') = i \sum_{mn} G_{mn}(t, t') W_{imjn}(t, t')$$


GF2 Closure: Second-order approximation in the bare Coulomb interactions

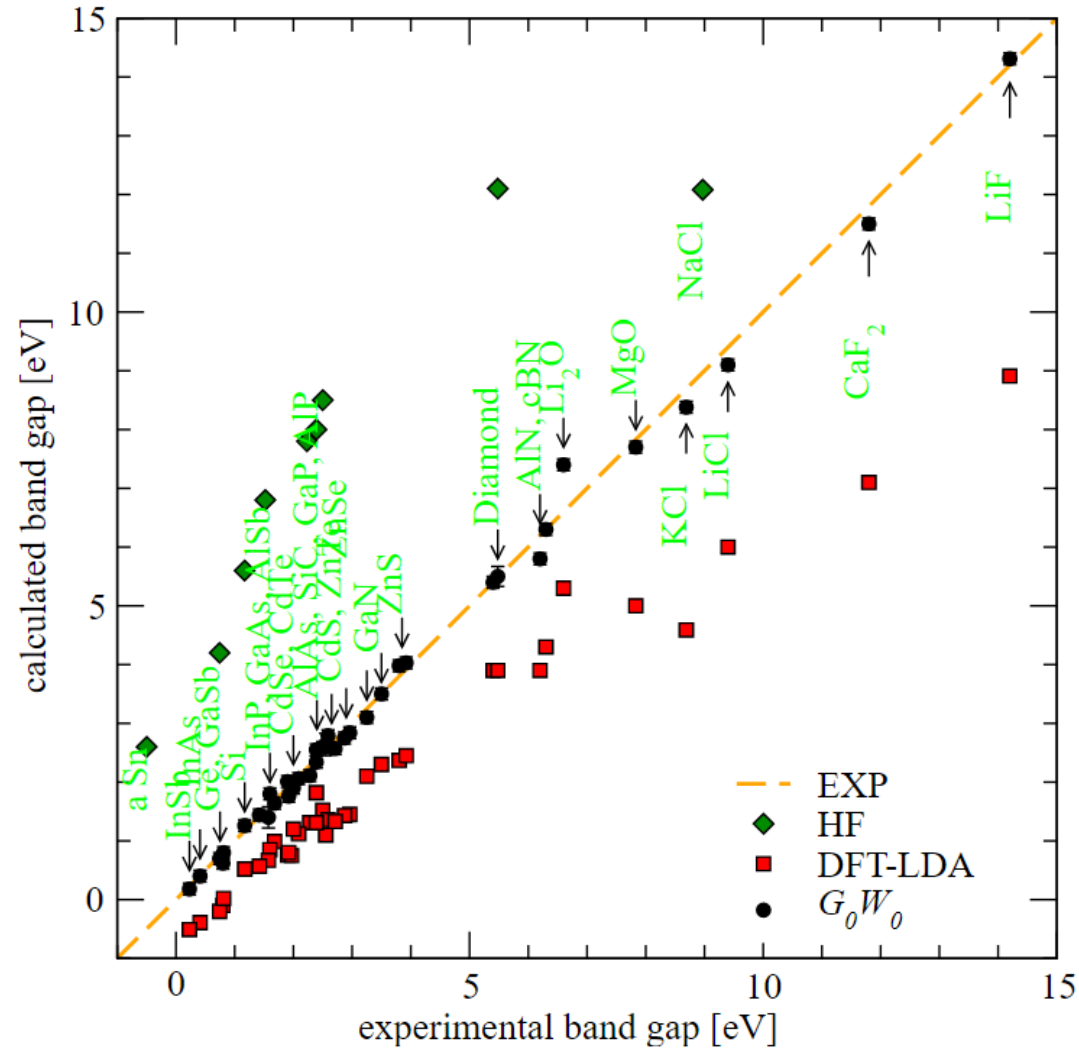
$$\Sigma^{\text{GF2}} = \text{Hartree-Fock} + \text{GW diagram} + \text{Second-order diagrams}$$

$$\Sigma_{ij}(t, t') = \sum_{mnklqp} G_{mn}(t, t') G_{kl}(t, t') G_{qp}(t', t) v_{impk} [2v_{jnql} - v_{jlqn}]$$

Electron Repulsion Integrals



GW in practice: semiconductor band gaps



Possibility to explore other approximations...

Lars Hedin 1999 J. Phys.: Condens. Matter 11 R489 + Others